

Research interests: Reduced-order modeling, data-driven modeling, machine learning, modal decompositions, transport-dominated physics, traveling waves, fluid dynamics, social good applications of data science

Education

University of Washington

2016-2021

Ph.D. in Mechanical Engineering, M.S. 2018

Seattle, WA

Thesis: Data-Driven Reduced-Order Modeling for Symmetrical Systems

Advisors: Steven Brunton (Mechanical Engineering) and J. Nathan Kutz (Applied Mathematics)

Seattle University

2012-2016

B.S. in Mechanical Engineering, Minor in Mathematics

Seattle, WA

Senior Design project with Ingersoll Rand, resulting in patent

Research advised by John Carter (Mathematics), supported by Clare Boothe Luce Foundation Undergraduate Research Award

Teaching

Instructor

DATA 5310	Data Visualization	F23, F22
DATA 5901	Capstone I	W26 ² , W25, W24, W23
DATA 5902	Capstone II	S26, S25, S24, S23
DATA 5322	Statistical Machine Learning II	S26*, S25 ² , S24, S23, S22
DATA 3310	Data Visualization	W26, W24, F22
MATH 1335	Calculus II	W22 ² , F21 ²

Bold face indicates a new course development, * indicates online modality, ² indicates two concurrent sections

Capstone Advisor

The Climate Pledge Recruitment Propensity	MSDS 26.1
Amazon Worldwide Sustainability	
Planning and Mission Effectiveness Demographic Data History	MSDS 25.2
Archdiocese of Seattle	
A Modern Approach to Pacific Salmon Research: An Online Dashboard	MSDS 25.5
NOAA Fisheries	
Meta-Analysis of Elwha River Water Temperature Regime	MSDS 24.3
NOAA Fisheries, Fish Ecology Division	
Washington State Craft Brewing Industry Snapshot	MSDS 24.5
Washington Brewers Guild	
Donor Database Analysis and Recommendations	MSDS 23.1
Camp Korey	
A Data Curation System and Predictive Model for Unconfined Concrete Strength	MSDS 23.3
DeSimone Consulting Engineers	

Publications

In Progress

The Geography of Nobel Prize Selection

CM Topaz, A Kurongonayini, B Ptak, A Fluehr, **A Mendible**, R Roca, N Rodríguez, L Xian, R Schwartz, J Higdon
Preprint DOI: 10.5281/zenodo.19262176

Quant Sci Studies

Under review 2026

Evaluating Remote Sensing and Machine Learning Methods for Predicting Understory Biomass in Post-Harvest Forests

SE Dixon, RJ McGaughey, **A Mendible**, BT Bormann, C Bobsin and GJ Ettl
Preprint DOI: 10.20944/preprints202607.0815.v1

In progress

2026

Peer-Reviewed Journal Publications

Community-Driven Data Science Practices

A Barreda II, C Díaz Eaton, S Hansen, JE Hibdon, L Gordon, R Greenwald, M José Gutiérrez Paz, K Ince, C Kelling, **A Mendible**, J Mercado, Victor Piercey, Bianca Thompson
DOI: 10.1007/s44007-026-00224-x

La Matematica

2026

Data collection and analysis for small-town policing: challenges and recommendations

C Kelling, A Haensch, **A Mendible**, S Brooks, A Wiedemann, M Aminian, W Hasty, J Higdon
DOI: 10.1080/2330443X.2024.2379270

Statistics and Public Policy

2024

Data science and social justice in the mathematics community

Q Jones, AR Vindas Meléndes, **A Mendible**, M Aminian, HZ Brooks, N Alexander, C Díaz Eaton, P Chodrow
DOI:10.1090/noti2773

Notices of the Am Math Soc

2023

Data-driven modeling of two-dimensional detonation wave fronts

A Mendible, W Lowrie, SL Brunton, JN Kutz
DOI: 10.1016/j.wavemoti.2021.102879

Wave Motion

2022

Data-driven modeling of rotating detonation waves

A Mendible, J Koch, H Lange, SL Brunton, JN Kutz
DOI: 10.1103/PhysRevFluids.6.050507

Phys Rev Fluids

2021

Dimensionality reduction and reduced-order modeling for traveling wave physics

A Mendible, SL Brunton, AY Aravkin, W Lowrie, JN Kutz
DOI: 10.1007/s00162-020-00529-9

Theor Comput Fluid Dyn

2020

Reduced strain mechanochemical activation onset in microstructured materials

JJ Schwartz, R Behrou, B Cao, M Bassford, **A Mendible**, C Shaeffer, AJ Boydston, N Boechler
DOI: 10.1039/C9PY01875A

Polym Chem

2020

Randomized nonnegative matrix factorization

NB Erichson, **A Mendible**, S Wihlbom, JN Kutz
DOI: 10.1016/j.patrec.2018.01.007

Pattern Recognit Lett

2018

Professional & Community-Facing Articles

Technical Appendix to *Connecting the Dots: A Preliminary Network Analysis of Complaints about Police in Minneapolis*

M Wang, **A Mendible**, C Kelling

DSPACE Blog

2026

Connecting the Dots: A Preliminary Network Analysis of Complaints about Police in Minneapolis

C Kelling, **A Mendible**, M Wang, A Wiedemann, T Marrinan, E Pederson

DSPACE Blog

2025

Counting on Justice: Bridging Quantitative Research and Community Empowerment

A Mendible, C Kelling

MAA Focus

2024

Quantitative Justice: Using Data Science for Good

A Mendible

MAA Math Values

2024

Getting Away From One Right Answer

A Mendible

COMAP Consortium Newsletter

2022

Small Town Police Accountability Lab
The Geography of Nobel Prize Selection

[Github](#)
[Github](#)

Patents

Air Flow Hour Meter

[US Patent 10,713,858](#)

A Mendible, S Heard, T Tran, A Hardesty

2017

Presentations

Data Science Tools for Community-Driven Police Accountability

[Jan 2025](#)

NAM-SIAM-AMS Special Session on Quantitative Justice, Joint Mathematical Meetings

[Seattle, WA](#)

Data Science Tools for Community-Driven Police Accountability

[Jul 2024](#)

Quantitative Justice through Applied Data Science Minisymposium, SIAM Annual Meeting

[Spokane, WA](#)

Panel: Decoding Ethics: Perspectives on Responsible Data Science

[May 2024](#)

Women In Data Science Puget Sound

[Seattle, WA](#)

Small Town Police Accountability: Case Studies and Opportunities in Data Science

[Jan 2024](#)

NAM-SIAM-AMS Special Session on Quantitative Justice, Joint Mathematical Meetings

[San Francisco, CA](#)

Data Science for Justice: Small Town Policing

[Nov 2023](#)

QSIDE Datathon4Justice, Opening Plenary

[Virtual](#)

Small Town Police Accountability: A Data Science Toolkit

[Jul 2023](#)

SciPy: Scientific Computing with Python

[Austin, TX](#)

Using Data Science for Social Justice

[Mar 2023](#)

Annual Meeting of the PNW Section of the Mathematical Association of America

[Newberg, OR](#)

QSIDE's STOPA (Small Town Policing Accountability) Research Lab

[Oct 2022](#)

QSIDE Datathon4Justice, Opening Plenary

[Virtual](#)

Meet Your Data Science Faculty

[Oct 2022](#)

Seattle University Mathematics Seminar

[Seattle, WA](#)

The Role of QSIDE's STOPA Research Lab in the Field of Criminal Justice

[Apr 2022](#)

QSIDE Data4Justice Conference

[Virtual](#)

Reduced-Order Modeling

[Mar 2022](#)

Pacific Lutheran University Mathematics Seminar

[Virtual](#)

Data-driven Modeling of Shock Waves in Rotating Detonation Engines

[Mar 2021](#)

SIAM Computational Science and Engineering

[Virtual](#)

Data-driven Modeling of Traveling Waves

[Dec 2020](#)

University of Delaware SIAM Seminar Series

[Virtual](#)

Data-driven Modeling of Traveling Waves

[2020](#)

Steve Brunton channel video abstract (14k views)

[YouTube](#)

Data-driven Modeling of Detonation Wave Interactions in Rotating Detonation Engines

[Nov 2020](#)

73rd Annual Meeting of the APS Division of Fluid Dynamics

[Virtual](#)

Dimensionality Reduction and Reduced Order Modeling for Traveling Wave Physics

[Nov 2019](#)

72nd Annual Meeting of the APS Division of Fluid Dynamics

[Seattle, WA](#)

Poster: The Space-Time Problem for Model Reduction of Traveling Waves

[Jun 2019](#)

DTRA Technical Program Review

[Fairfax, VA](#)

The Space-time Problem with Model Reduction for Traveling Waves

[Mar 2019](#)

SIAM Computational Science and Engineering

[Spokane, WA](#)

Scholarly Development

Grant Funding

Comparative Police Complaint Network Analysis: Supporting Undergraduate Researchers in Community-Engaged Data Science	<i>\$1,000</i>
American Statistical Association	2026
Bridging Data & Community Micro Grant	
Comparative Police Complaint Network Analysis	<i>\$7,000</i>
University of Minnesota Institute for Advanced Study	2026
Research and Creative Collaborative with Reinvestigation Workgroup, Co-PI	
Optimization of Cryoprotective Agents for Single-Mode Electromagnetic Resonant (SMER) Rewarming Technology to Enable Long-Term Organ Preservation	<i>\$50,000</i>
Seattle U College of Science and Engineering	2023
Multidisciplinary Research Program Award with Shen Ren (Mechanical Engineering)	
Small Town Police Accountability Lab	<i>Workshop hosting</i>
Banff International Research Station for Mathematical Innovation and Discovery	2022
Proposal accepted	

Pending and Attempted Funding Applications

Fostering Opportunity, Readiness and Growth for Entry (FORGE) Into STEM Careers	<i>\$2,000,000</i>
National Science Foundation	2026
Scholarships in Science, Technology, Engineering, and Mathematics (S-STEM) Program, Co-PI	
Development of Integrated Data Science Tools Towards Understanding of Perceptions of Policing Through Public Data	<i>\$364,054</i>
National Institute of Justice	2024
Measurement of Community Perceptions, PI	

Workshops

Reproducible Machine Learning Workflows for Scientists	<i>Aug 2025</i>
US Research Software Sustainability Institute, Funded Participant	Madison, WI
Community Engaged Learning Faculty Fellowship Workshop	<i>Sep 2023</i>
Seattle U Center for Community Engagement, Funded Participant	Seattle, WA
Educating at the Intersection of Data Science and Social Justice Workshop	<i>Jul 2023</i>
Institute for Computational and Experimental Research in Mathematics, Brown University, Funded Participant	Providence, RI
Community Engaged Learning Workshop Series	<i>May 2023</i>
Seattle U Center for Community Engagement	Seattle, WA
Data Science and Social Justice: Networks, Policy, and Education Workshop	<i>Jun-Jul 2022</i>
Institute for Computational and Experimental Research in Mathematics, Brown University, Funded Participant	Providence, RI
Project NExT Fellow	<i>2021-2022</i>
Mathematical Association of America	Virtual
Preparing for Careers in Teaching Statistics and Data Science Workshop	<i>Aug 2021</i>
Section on Statistics and Data Science Education of the American Statistical Association	Virtual
EPSRC Summer School: Modal Decompositions in Fluid Mechanics	<i>Aug 2019</i>
Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Funded Participant	Cambridge, UK

Research Students

Jesse Loi	<i>AY25-26</i>
Data Accessibility for Public Accountability	MS Data Science
supported by SU Technology Ethics Initiative Student-Scholar Program, poster presentations at SRRCON 2026 and SciPy 2026	
David Stanko and Mark Daza	<i>FQ25</i>
Applications of Language Models for Complaint Data: NER and LLMs	MS Data Science

Emma Oriol

Historical Redlining and Modern Gentrification

AY23-24

MS Data Science

Alexander Churhill

Historical Redlining and Housing Quality in Seattle

AY23-24

MS Data Science

Chang He

Historical Redlining and Cycling Accessibility

AY23-24

MS Data Science

Malia Cortez

Historical Redlining and Population Flows

AY23-24

MS Data Science

Service**Academic Service****Capstone Director, MS Data Science**

Select industry sponsors, scope projects, recruitment and information events

2022-present

Semester Transition Coordinator, Data Science

Lead transition from quarters to semesters for data science programs

2025-present

MS Data Science Online Course Creation

Create material for four asynchronous courses for online MSDS program

2025-present

Chile Study Abroad Trip Co-Lead

Assisted in supervising spring break student trip for Mathematical Models of Near-Shore Phenomena course

2024, 2026

Math Department Recruitment, Inclusion, Diversity, Equity and Retention Committee

Member, led climate survey creation and distribution

2021-present

QSIDE Institute Consortium Liaison

Lead event coordination, advertisement at SU, and student involvement

2021-present

Bannan Building Safety Committee**Math Department Community Building**

Organize end-of-year student celebration for graduating MSDS cohort

2023, 2024, 2025, 2026

Organize weekly (AY 22-23) or quarterly (AY 23-24) activities and events to build departmental community

2022-2024

Math Department Website Review

2025

MS Data Science Student Admissions Review

Review ~250 student applications per year, recommended admissions decisions

2022-2025

Data Science Faculty Hiring Committee

Hosted 6 all-day candidates, 2 hires

2025

Reviewed 54 candidate applications, 14 screening interviews, hosted 4 all-day candidates, 2 hires

2023

Reviewed ~9 candidate applications, 9 screening interviews, hosted 2 all-day candidates, position unfilled

2022

Professional Service**QSIDE Data Science, Police Accountability and Community Engagement Research Lab**

Co-Director, lead meetings and organize research activities

2021-present

SciPy Conference

Program Committee, organize topics and oversee tracks conference-wide

2025, 2026

Track Chair: Human Networks, Social Sciences, and Economics

2024

DSPACE Datathon

Organizer, led two hybrid datathons held between three campuses

2026

Women In Data Science Puget Sound

Invited mentor for Mentorship Night for career-seeking women

2024

QSIDE Datathon4Justice

Co-Organizer

2023

Team Lead

2021, 2022, 2023

Journal Referee

Physical Review Fluids

La Matematica

Community Service

Mechanical Engineering Students Against Racism , UW Mechanical Engineering	2020-2021
Crafters Against CoVID-19	2020
Engineering Discovery Days , UW College of Engineering	2016, 2019
Bike & Build Drift West	2017
UW Society of Women Engineers , Elementary school outreach visits	2016

Honors & Awards

Community Engaged Learning Faculty Fellowship , Seattle U Center for Community Engagement	AY 23-24
Supervisor of the Year , Seattle University Student Employment Shine Award	2023
Project NExT Fellow , Mathematical Association of America	2021
Graduate Diversity Fellowship , UW Graduate Opportunities and Minority Achievement Program, Full Support	2016, 2020
Steve and Lynn Pratt Fellowship , UW Mechanical Engineering	2017, 2020
Wynne Alexander Guy Spirit of the Mathematics Department Award , SU Mathematics	2019
Wei Li and Dongmei Chen Endowed Fellowship , UW Mechanical Engineering	2017
Brian and Mary Fabien Endowed Fellowship , UW Mechanical Engineering	2016, 2017
Undergraduate Research Award , Clare Boothe Luce Foundation	2014

Additional Experience

Clear Motion (formerly Levant Power)	June - Aug 2016
Controls Engineering Intern	Woburn, MA
National Renewable Energy Laboratory	June - Aug 2015
Science Undergraduate Laboratory Internship	Boulder, CO
Network for Earthquake Engineering Simulation	June - Aug 2014
OH Hinsdale Wave Research Laboratory REU	Corvallis, OR